

Part A: Property Price Prediction

1. Overview

This project focuses on predicting property prices in various districts of California using several district-level features. By building a predictive model, we aim to identify key variables that influence housing prices and improve the accuracy of house value predictions. The project will specifically utilize **simple linear regression** and **multiple linear regression** to address this regression task, ensuring proper data handling and evaluation of the models.

2. Problem Statement

The objective is to predict the **median house value** in California districts based on features such as income, the number of rooms, geographical location, and proximity to the ocean. Given the dataset, we will develop regression models, evaluate their performance, and determine which model provides the best balance between predictive accuracy and interpretability.

3. Dataset Information

Dataset:  **Data_file**

The information related to variables given can be found in the [Data Information.pdf](#).

4. Deliverables

- Exploratory Data Analysis (EDA) with visualizations and summary statistics.
 - Data Preprocessing, including handling missing values and encoding categorical variables.
 - Model development using:
 - Simple Linear Regression
 - Multiple Linear Regression
 - Evaluation of the models using relevant metrics (Confusion Matrix, Precision, Recall, F1-Score, etc.) where applicable.
 - A final, well-documented notebook detailing all steps, insights, and the final model selection.
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5. Success Criteria

- The model should have a high degree of accuracy and balance with interpretability.

- The evaluation metrics such as MSE, RMSE and R Square will be used to measure the model's performance.
 - Ensure proper documentation of all steps, and present visualizations that help explain the data and model outcomes.
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6. Guidelines

- Make sure to split your data into training and testing sets to avoid overfitting.
 - Tune the hyperparameters of your models to improve performance.
 - Report all the steps taken in the data preprocessing, modeling, and evaluation phases.
 - Provide a final model that balances accuracy with interpretability.
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7. Tools Required

- Python (with libraries such as pandas, scikit-learn, matplotlib, seaborn, etc.)
- Jupyter Notebook or any IDE suitable for running Python code

Part B: Customer Churn Prediction

1. Overview

Customer churn, or customer attrition, refers to when a customer ceases their relationship with a company or service provider. In today's highly competitive business environment, retaining customers is a critical factor for long-term success. Predicting customer churn can help organizations take proactive steps to retain customers, thus minimizing revenue loss. This project aims to build a machine learning model that can predict whether a customer will churn based on their demographic, account, and service-related data.

2. Problem Statement

The goal of this project is to develop a classification model that predicts whether a customer will churn. Using demographic data (such as gender, senior citizen status, and tenure), along with information about the services they use (such as internet service, phone service, and online security), we will attempt to build a model that helps the company identify customers who are at a high risk of churning.

By predicting customer churn, the company can proactively design retention strategies to keep these customers, thereby improving customer satisfaction and reducing financial loss.

3. Dataset Information

Dataset:  Data_file

The information related to variables given can be found in the [Data Information.pdf](#).

4. Deliverables

- A data exploration and preprocessing notebook or report that analyzes the dataset, handles missing values, and prepares the data for modeling.
- A machine learning model capable of predicting customer churn.
- An evaluation of model performance using appropriate metrics (such as accuracy, precision, recall, F1 score, etc.).
- A final report explaining the insights gained from the model and the business implications of customer churn prediction.

5. Success Criteria

The success of the project will be determined by the following:

- Proper interpretation of the model's output, providing actionable insights to reduce customer churn.
- Get the predictions for the new data.

6. Guidelines

- Make sure to split your data into training and testing sets to avoid overfitting.
 - Tune the hyperparameters of your models to improve performance.
 - Report all the steps taken in the data preprocessing, modeling, and evaluation phases.
 - Provide a final model that balances accuracy with interpretability.
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7. Tools Required

- Python (with libraries such as pandas, scikit-learn, matplotlib, seaborn, etc.)
- Jupyter Notebook or any IDE suitable for running Python code

Part C: Early Disease Detection

1. Overview

Cardiovascular diseases (CVDs), including heart disease, are the leading cause of death worldwide. Early detection of heart disease is critical for preventing serious health outcomes and improving the quality of life for patients. With the increasing availability of medical data, machine learning models can be used to predict whether a patient is likely to develop heart disease based on certain health indicators.

In this project, you will build a classification model to predict whether an individual is likely to have heart disease or not. The dataset provided includes various health and demographic factors such as age, blood pressure, cholesterol levels, and lifestyle habits (e.g., smoking

and alcohol consumption). The goal is to train a model to identify which individuals have heart disease based on these features.

2. Problem Statement

You are provided with a dataset that contains health-related information about individuals. Your task is to develop a machine learning model that can predict the presence of heart disease based on the provided features. The target variable in the dataset is "disease," which indicates whether a person has heart disease (1) or not (0).

You need to perform the following tasks:

Data Exploration and Preprocessing: Understand the dataset, handle missing values, perform feature engineering if necessary, and prepare the data for model training.

Model Development: Train a classification model to predict the presence of heart disease using the features provided in the dataset.

Model Evaluation: Evaluate the model's performance using appropriate classification metrics such as accuracy, precision, recall, and F1-score. Identify the best-performing model based on these metrics.

Insights and Reporting: Analyze the results and provide insights into which factors are the most significant predictors of heart disease.

3. Dataset Information

Dataset:  **Data_file**

The information related to variables given can be found in the [Data Information.pdf](#).

4. Deliverables

Exploratory Data Analysis (EDA): Analyze the dataset to understand the distribution of the variables, check for missing data, and identify any relationships or patterns between the features and the target variable (disease).

Data Preprocessing: Handle missing or erroneous values, normalize/standardize data if necessary, and perform feature engineering if required.

Model Development: Train various classification models (e.g., Logistic Regression, Decision Trees, SVM, etc.) and compare their performance.

Model Evaluation: Evaluate your models using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score

Insights and Conclusion: Based on your model and analysis, provide insights into the factors that are most predictive of heart disease and make recommendations on how to improve heart disease prediction models.

5. Success Criteria

A well-documented Jupyter notebook or code file showcasing the entire workflow from data exploration to model evaluation.

Insights derived from the data and model results that provide a better understanding of the risk factors associated with heart disease.

6. Guidelines

Make sure to split your data into training and testing sets to avoid overfitting.

Tune the hyperparameters of your models to improve performance.

Report all the steps taken in the data preprocessing, modeling, and evaluation phases.

Provide a final model that balances accuracy with interpretability.

7.Tools Required

Python (with libraries such as pandas, scikit-learn, matplotlib, seaborn, etc.)

Jupyter Notebook or any IDE suitable for running Python code